ClipSAT : Differential Equations First/Second Order, Laplace, Systems & S

Unit 5.01: Variation of Parameters & Greens Kernel

Name: _____

Q1. Find a particular solution of y''-y=2 using variation of parameters.

[ ] A: 2
[ ] B: -2
[ ] C: -1
[ ] D: -tfrac12

Q2. Find a particular solution of y''-4y'+4y=6 using variation of parameters.

[ ] A: -tfrac32
[ ] B: 3
[ ] C: tfrac32
[ ] D: 6

Q3. Find a particular solution of y''+4y=8 using variation of parameters.

[ ] A: 2
[ ] B: -2
[ ] C: 4
[ ] D: 8

Q4. Find a particular solution of y''+y=cot x for 0<x<pi using variation of parameters.

[ ] A: sin xln|csc x+cot x|
[ ] B: -cos xln|csc x+cot x|
[ ] C: -sin xln|sec x+tan x|
[ ] D: -sin xln|csc x+cot x|

Q5. Find a particular solution of y''-4y=e^{2x} using variation of parameters.

[ ] A: Bigtfrac{x}{4}+tfrac{1}{16}Bige^{2x}
[ ] B: Bigtfrac{x}{4}-tfrac{1}{16}Bige^{2x}
[ ] C: Bigtfrac{x}{2}-tfrac{1}{16}Bige^{2x}
[ ] D: -Bigtfrac{x}{4}-tfrac{1}{16}Bige^{2x

Q6. Find a particular solution of y''+4y=csc2x for 0<x<pi/2 using variation of parameters.

[ ] A: -tfrac{x}{2}cos2x+tfrac14sin2xln|sin
[ ] B: tfrac{x}{2}cos2x+tfrac14sin2xln|sin2
[ ] C: -tfrac{x}{2}cos2x-tfrac14sin2xln|sin
[ ] D: -tfrac{x}{4}cos2x+tfrac14sin2xln|sin

Q7. Find a particular solution of y''-6y'+9y=dfrac{e^{3x}}{x^2} for x>0 using variation of parameters.

[ ] A: e^{3x}ln x+1
[ ] B: -e^{3x}ln x-1
[ ] C: -e^{3x}ln x+1
[ ] D: -e^{3x}ln x

Q8. Find a particular solution of y''+4y'+3y=e^{-x} using variation of parameters.

[ ] A: Bigtfrac{x}{2}+tfrac14Bige^{-x}
[ ] B: x-tfrac14e^{-x}
[ ] C: -Bigtfrac{x}{2}-tfrac14Bige^{-x}
[ ] D: Bigtfrac{x}{2}-tfrac14Bige^{-x}